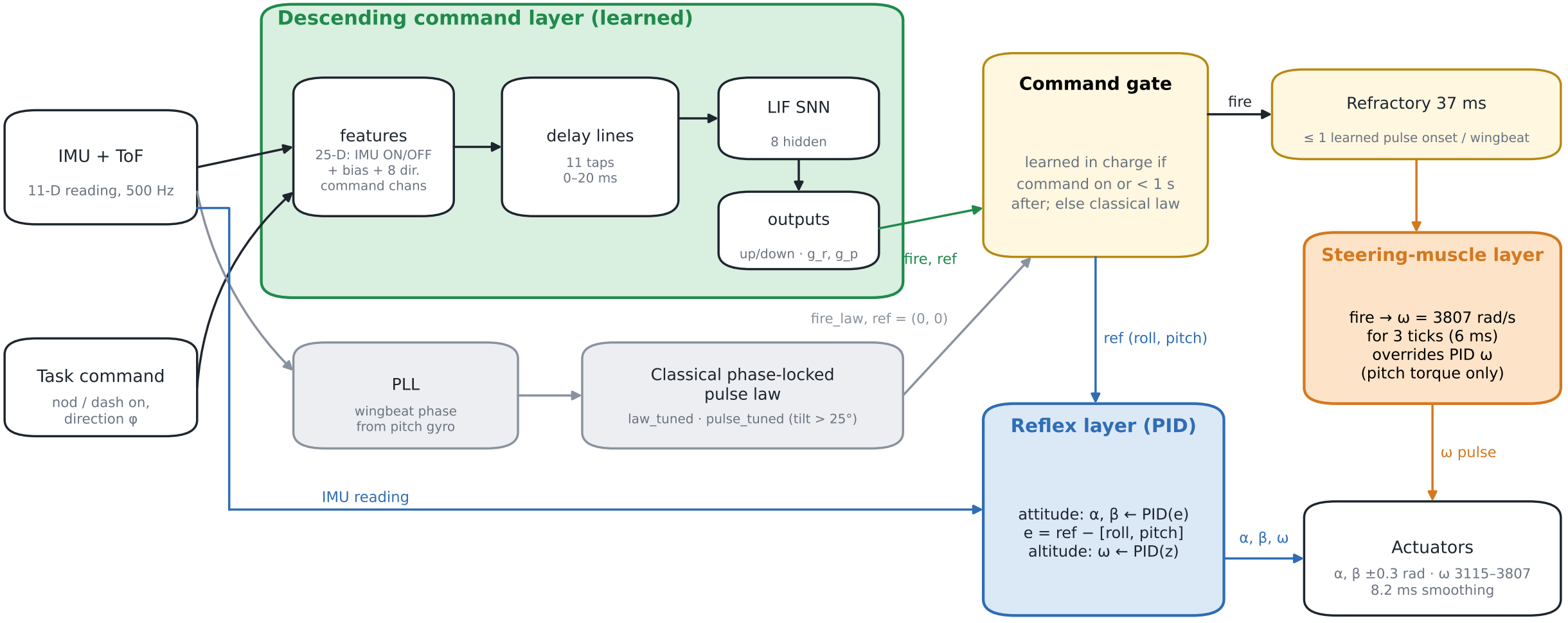
Bio-inspired hierarchical controller, plan v6: nod in 4 / dash in 8 directions (one 2 ms tick)



Fixed interface: sensors, actuators, update rate and limits are those of the original environment. Pulse direction (pitch up / down) is set only by its timing within the wingbeat (≈135° up, ≈315° down); the single motor gives no roll torque, so roll manoeuvres go through the roll set-point (g_r → PID → gimbal).